



LeetCode 1413 — Minimum Value to Get Positive Step by Step Sum

1. Problem Title & Link

Title: LeetCode 1413: Minimum Value to Get Positive Step by Step Sum

Link: <https://leetcode.com/problems/minimum-value-to-get-positive-step-by-step-sum/>

2. Problem Statement (Short Summary)

Given an integer array `nums`, choose a starting value `startValue`.

While traversing the array:

$\text{currentSum} = \text{startValue} + \text{nums}[0] + \text{nums}[1] + \dots$

At every step, the running sum must remain:

≥ 1

Return the minimum starting value that satisfies this condition.

3. Examples (Input → Output)

Example 1

Input:

`nums = [-3,2,-3,4,2]`

Output:

5

Explanation:

Start with 5

$5 + (-3) = 2$

$2 + 2 = 4$

$4 + (-3) = 1$

$1 + 4 = 5$

$5 + 2 = 7$

Never goes below 1.

Example 2

Input:

`nums = [1,2]`

Output:

1

Explanation:

$1 \rightarrow 2 \rightarrow 4$

Always positive.



Example 3

Input:

nums = [1,-2,-3]

Output:

5

Explanation:

$5 \rightarrow 6 \rightarrow 4 \rightarrow 1$

Minimum valid value.

4. Constraints

$1 \leq \text{nums.length} \leq 100$

$-100 \leq \text{nums}[i] \leq 100$

Important:

Need running sum ≥ 1 at every step.

Not just the final sum.

5. Core Concept (Pattern / Topic)

Prefix Sum

This is one of the easiest and most important Prefix Sum intuition problems.

6. Thought Process (Step-by-Step Explanation)

Brute Force Approach

Try:

startValue = 1

Simulate.

If it fails:

startValue = 2

Try again.

Continue until valid.

Example:

nums = [-3,2,-3,4,2]

Try:

1 ❌

2 ❌

3 ❌



4 ❌

5 ✅

Works but inefficient.

Optimized Thinking

Suppose:

nums = [-3,2,-3,4,2]

Compute prefix sums.

-3

-1

-4

0

2

The smallest prefix sum is:

-4

If starting value is:

x

Then minimum running sum becomes:

$x + (-4)$

We need:

$x + (-4) \geq 1$

Therefore:

$x \geq 5$

Answer:

5

No trial and error needed.

7. Visual / Intuition Diagram (ASCII)

```
nums
[-3, 2, -3, 4, 2]
Prefix Sums
-3
-1
-4
0
2
Minimum Prefix Sum
-4
```



Need:
 $\text{startValue} + (-4) \geq 1$
Therefore:
 $\text{startValue} \geq 5$
Answer:
5

8. Pseudocode

```
runningSum = 0

minimumPrefix = 0

for each number

    runningSum += number

    minimumPrefix =
        min(minimumPrefix, runningSum)

return 1 - minimumPrefix
```

9. Code Implementation

Python

```
class Solution:
    def minStartValue(self, nums):

        runningSum = 0
        minPrefix = 0

        for num in nums:

            runningSum += num

            minPrefix = min(minPrefix, runningSum)

        return 1 - minPrefix
```

Java

```
class Solution {

    public int minStartValue(int[] nums) {

        int runningSum = 0;
        int minPrefix = 0;

    }
```



```
        for(int num : nums) {  
            runningSum += num;  
            minPrefix = Math.min(minPrefix, runningSum);  
        }  
        return 1 - minPrefix;  
    }  
}
```

10. Time & Space Complexity

Time Complexity

$O(n)$

Single traversal.

Space Complexity

$O(1)$

Only variables used.

11. Common Mistakes / Edge Cases

Mistake 1

Checking final sum only.

Wrong:

Final sum > 0

The problem requires:

Every intermediate sum ≥ 1

Mistake 2

Tracking maximum prefix sum.

Need:

Minimum prefix sum

Mistake 3

Returning:

$-\text{minPrefix}$

Correct formula:

$1 - \text{minPrefix}$



12. Detailed Dry Run

Input:

nums = [-3,2,-3,4,2]

Num	Running Sum	Min Prefix
-3	-3	-3
2	-1	-3
-3	-4	-4
4	0	-4
2	2	-4

Minimum Prefix:

-4

Answer:

1 - (-4)

= 5

Output:

5

13. Common Use Cases (Real-Life / Interview)

Account Balance Tracking

Suppose:

Transactions:

-300

+200

-300

+400

Find minimum starting balance so the account never becomes negative.

Same idea.

Other examples:

Battery charge monitoring

Inventory management

Cash flow analysis

Resource allocation



14. Common Traps (Important!)

Trap 1

Using brute force simulation.

Works but unnecessary.

Trap 2

Forgetting that:

Minimum Prefix Sum

determines the answer.

Trap 3

Not initializing:

`minPrefix = 0`

Very important.

15. Builds To (Related LeetCode Problems)

LC 1413 Minimum Start Value



LC 303 Range Sum Query



LC 724 Find Pivot Index



LC 560 Subarray Sum Equals K



LC 525 Contiguous Array

Prefix Sum Roadmap.

16. Alternate Approaches + Comparison

Approach	Time	Space	Notes
Brute Force	$O(\text{answer} \times n)$	$O(1)$	Slow
Simulation + Increment	$O(\text{answer} \times n)$	$O(1)$	Inefficient
Prefix Sum	$O(n)$	$O(1)$	Optimal

Approach 1: Brute Force

Try:

1

2



3

4

...

until valid.

Very slow.

Approach 2: Prefix Sum

Track:

minimum prefix sum

Compute answer directly.

Best solution.

17. Why This Solution Works (Short Intuition)

The lowest point reached by the running sum determines how much extra starting value we need.

If the minimum prefix sum is:

-4

then we need:

4 more units

plus:

1

to stay positive.

Therefore:

answer = 1 - minimumPrefix

18. Variations / Follow-Up Questions

Follow-Up 1

Keep running sum always greater than 10.

Answer:

10 - minimumPrefix + 1

Follow-Up 2

Find maximum deficit reached.

Equivalent to finding:

minimum prefix sum

Follow-Up 3

What if updates arrive continuously?



Leads to dynamic prefix sum problems.

Follow-Up 4

What if multiple queries are asked?

Leads to prefix sum arrays and segment trees.

Pattern Learned

Prefix Sum

Running Sum



Track Minimum Prefix



Find Worst Dip



Compute Required Starting Value